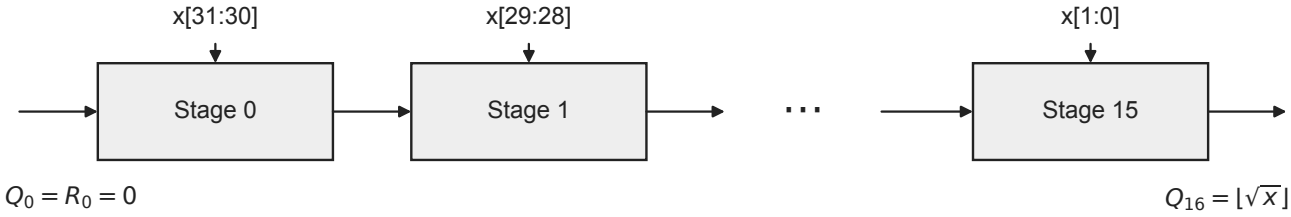
32-bit radicand: 16 input pairs, processed most significant first



One combinational stage

$$T = 4R_k + b_k, \quad D = 4Q_k + 1$$

If $T \geq D$: $(Q_{k+1}, R_{k+1}) = (2Q_k + 1, T - D)$

If $T < D$: $(Q_{k+1}, R_{k+1}) = (2Q_k, T)$

Invariant: prefix $= Q_k^2 + R_k$

Example: x = 20

Pair	T	D	Q next	R next
01	1	1	1	0
01	1	5	2	1
00	4	9	4	4

$4^2 + 4 = 20$; returned root = 4